

When Is an Inverse a Function? - Practice

Algebra II | Unit 1 | Lesson 12 Practice | TEK A2.2.C

Objective: I can decide whether an inverse is a function, and restrict a domain when it is not.

Section 1 — Recall

1. Given $f(x) = x^2$, evaluate $f(3)$ and $f(-3)$. What do you notice?

$f(3) =$ _____ $f(-3) =$ _____

2. Is $f(x) = x^2$ one-to-one? Why or why not?

Section 2 — Apply

3. Restrict $f(x) = x^2$ to $x \leq 0$. What is $f^{-1}(x)$ under this restriction?

$f^{-1}(x) =$ _____

4. $g(x) = x^2$ is restricted to $x \geq 0$. Give the range of g^{-1} under this restriction.

Range: _____

Section 3 — Challenge

5. $f(x) = |x|$ restricted to $x \leq 0$. Find $f^{-1}(x)$ under this restriction, and verify with one test point.

$f^{-1}(x) =$ _____ Check: _____

6. True or false: every parabola ($ax^2 + bx + c$) needs a domain restriction for its inverse to be a function. Justify, or give a counterexample.

Section 4 — Explain It

7. A classmate says: "Any restriction on the domain of $f(x) = x^2$ will fix the one-to-one problem, as long as SOME numbers get excluded." Test the claim using the restriction $-3 \leq x \leq 3$. Is the classmate correct?
